It is important to remember that any time our circuit has a connection to ground that this must be common to the Arduino and every connected device or things will behave very erratically. What this means is that the ground pins of the circuit must somehow connect to the ground pins on the microcontroller. Likewise, any time we use a pin for an input that is not actually connected to anything, it is called a **floating pin** and we will get seemingly random and strange results as the pin picks up electrical interference and noise from the surrounding environment or from nearby circuits.

Project 4: Noisy Cricket

As a means to explore the uses of the Arduino's digital pins and to discuss the different digital functions, we are going to look at a new project: the Noisy Cricket. This project will make quite an annoying racket when left alone, but it will quickly hush up when someone gets near it—just like a real cricket. We will use a simple and inexpensive motion sensor as a trigger, although a tilt switch, mat switch, or any other simple on or off sensor will do. Wire the whole thing up and power it from a battery, then go find a hard-to-reach place for the cricket to call home and see how many people you can drive insane with it!

Hooking It Up

This circuit can be as simple or complex as you want to make it. We will go with the easy route, using an inexpensive, passive infrared (PIR)motion detector. We are somewhat partial to the ones from Parallax or Adafruit, listed in the appendix at the end of this book, for their ease of use and connectivity. A PIR motion detector is really good at detecting a change in the nearby ambient infrared temperature usually caused by the movement of warm-blooded creatures like us. It needs three connections: +5 VDC, GND, and OUT, however, double-check the sensor's technical specifications to make sure the correct connections are made to the right pins. The output pin is usually configurable and can provide a logic level `HIGH` when motion is detected and `LOW` otherwise. Alternatively, you could stick with the tilt sensor circuit from earlier or some other form of digital switch or sensor.

For the speaker, we are using a piezo buzzer, which uses a piezoelectric crystal coupled with a small diaphragm to make sound. Some piezo speakers are polarized and will have a side marked '+' that should be connected to the Arduino output pin. Instead of a piezo, a more conventional small, simple 8 ohm speaker like you would find in an electronic toy or alarm clock could also work with a 100 ohm series resistor placed between the output pin and the speaker. Figures 5-4 and 5-5 show how all of this goes together.